AutoML Experiment Report: classifying_smoker_status

AutoHealth

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Abstract

This report documents an automated machine learning workflow for smoker status classification from structured health examination data. The task is a clinically relevant binary classification problem: estimating whether an individual is a smoker based on routine demographic, biochemical, and vital-sign measurements. The final system combines a feature-selected LightGBM classifier with a lightweight snapshot ensemble and a Platt-scaling calibration layer. On the held-out test set, the calibrated model achieved an accuracy of 0.8191, ROC-AUC of 0.8970, precision of 0.7455, recall of 0.7703, and F1-score of 0.7577. Probability quality also improved after calibration, with Expected Calibration Error reduced from 0.0456 to 0.0278 and Negative Log-Likelihood reduced from 0.4021 to 0.3931. In addition, the ensemble dispersion signal provided a useful uncertainty proxy: incorrect predictions were, on average, about twice as uncertain as correct ones. Overall, the model is strong enough for screening or decision-support settings, though not yet ideal for fully automated use without human review in ambiguous cases.

1 Data Analysis

1.1 Dataset Overview

The dataset contains 44,553 training samples, 5,568 validation samples, and 5,571 test samples. The target variable is binary smoking status. The class ratio is stable across all three splits, which makes validation and test estimates easier to interpret and reduces the risk that performance differences are driven by sampling artifacts. Based on the upstream analysis, the non-smoker class accounts for roughly 63% of observations and the smoker class about 37%, so the task is moderately imbalanced rather than severely skewed.

1.2 Feature Characteristics

The raw table combines demographic information, anthropometric measurements, laboratory markers, and routine clinical observations. Several predictors have clear physiological relevance for smoking behavior. Hemoglobin, height, triglyceride, Gtp, and body-size-related variables all show meaningful separation between smokers and non-smokers. HDL and age move in the opposite direction and help improve discrimination by capturing metabolic and behavioral structure that differs across smoking groups.

One particularly important signal is the strong sex effect noted during exploratory analysis: the smoking prevalence among males is substantially higher than among females in this dataset. This does not invalidate the model, but it does mean that subgroup behavior and calibration deserve explicit attention in later evaluation. To strengthen metabolic information capture, an engineered triglyceride-to-HDL ratio was added as an interpretable composite feature.

1.3 Data Quality and Modeling Implications

The dataset is relatively clean. There are no major missing-value problems in the final modeling table, the constant column was removed, and the retained variables are well-suited to gradient-boosted tree methods. At the same time, exploratory analysis suggested strong correlation structure among cardiometabolic variables. This justified feature selection before final training, both to improve efficiency and to reduce redundancy in the learned decision function.

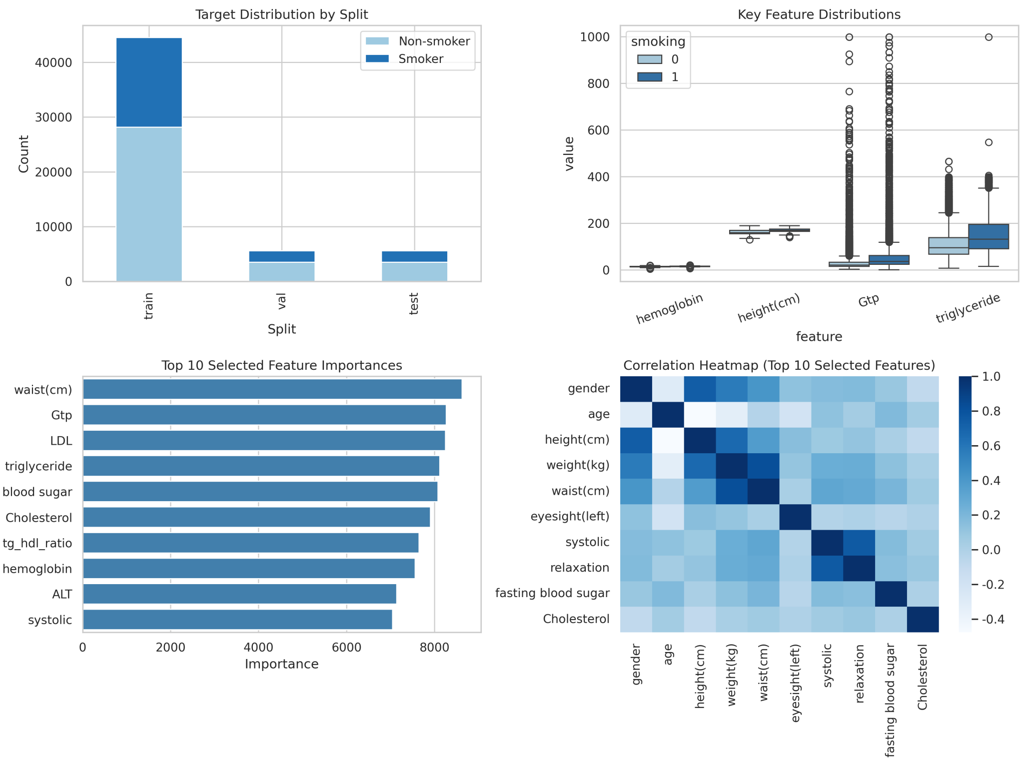

Figure 1: Summary of the exploratory analysis, including class distribution, key feature separation, feature importance, and correlation structure among selected variables. The figure illustrates both the signal strength of smoking-related biomarkers and the redundancy structure that motivated feature filtering.

2 Model Training

2.1 Preprocessing Pipeline

The preprocessing pipeline removed the identifier column and the constant oral-exam column, encoded gender numerically, created the triglyceride-to-HDL ratio, standardized numerical variables using training-set statistics only, and retained 20 final predictors after recursive feature elimination. This sequence was designed to avoid leakage while preserving the strongest clinically plausible signals.

2.2 Model Choice

The final predictive model is a LightGBM classifier. This is a sensible choice for medium-scale structured clinical data because it handles heterogeneous feature distributions well, captures non-linear effects and threshold behavior, and remains efficient enough for repeated validation and post-hoc analysis. Hyperparameters were tuned through a constrained Optuna search, after which the model was retrained with early stopping on validation binary logloss.

2.3 Snapshot Ensemble and Calibration

Instead of training multiple independent models, the pipeline used a lightweight snapshot strategy around the best boosting iteration. Predictions from nearby rounds were averaged to form a small ensemble, and the standard deviation across those predictions was used as an uncertainty score. A Platt-scaling layer was then fitted on validation probabilities to improve calibration. This is a pragmatic design: it adds some of the benefits of ensembling and uncertainty estimation without incurring the cost of maintaining many separate tree models.

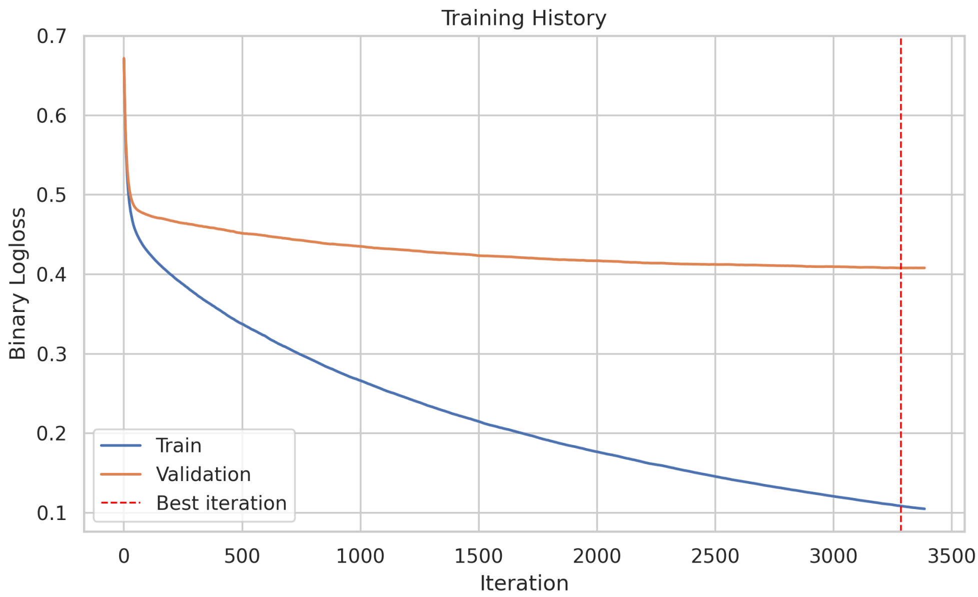

Figure 2: Training and validation binary logloss across boosting iterations. The curve shows stable optimization behavior and supports the use of early stopping to select the final ensemble center.

3 Model Performance

3.1 Overall Predictive Quality

Table 1: Test Set Metrics

Metric	Value
Accuracy	0.8191
ROC-AUC	0.8970
Precision	0.7455
Recall	0.7703
F1-score	0.7577
ECE	0.0278
Brier Score	0.1249

The most important headline result is the ROC-AUC of 0.8970, which indicates that the model has strong ranking ability: smokers are generally assigned higher predicted probabilities than non-smokers. Accuracy, precision, recall, and F1-score are also all balanced enough to suggest that the classifier is not simply exploiting the majority class. In particular, recall of 0.7703 means the model identifies a substantial fraction of smokers, while precision of 0.7455 indicates that positive predictions are not excessively noisy.

3.2 Calibration Quality

Calibration is a major strength of the final pipeline relative to an uncalibrated tree model. The Brier score improved from 0.1266 to 0.1249 after calibration, the Expected Calibration Error fell from 0.0456 to 0.0278, and the Negative Log-Likelihood also improved. These changes matter because in practical use, a predicted probability should be interpreted as a meaningful risk estimate rather than only as a ranking score. The post-calibration values are not perfect, but they are good enough to make the model more credible in decision-support scenarios.

3.3 Interpretation and Limits

The model is substantially better at ranking risk than at guaranteeing perfectly calibrated probabilities across every subgroup and operating threshold. That is typical in moderately imbalanced clinical-style tabular tasks. The current 0.5 decision threshold is a reasonable generic choice, but it is unlikely to be optimal for every deployment setting. If the downstream goal is to maximize smoker detection, a lower threshold could improve sensitivity at the cost of more false positives. If the goal is conservative flagging, the current operating point is defensible.

4 Uncertainty Analysis

4.1 Uncertainty Signal

Prediction standard deviation across snapshot members was used as the uncertainty score. This is a simple but effective approximation of model instability around the local optimum. In the final results, incorrect predictions had meaningfully higher uncertainty than correct predictions, which indicates that the ensemble spread is carrying real information about prediction difficulty rather than random noise.

4.2 Subgroup Perspective

The uncertainty analysis was also broken down by gender. While the report does not claim strict fairness guarantees, the subgroup plots provide a useful sanity check: they show whether uncertainty behaves similarly for male and female cohorts and whether one subgroup accumulates disproportionately ambiguous cases. This matters because sex is both a strong predictor and a potential source of shortcut learning in this dataset.

4.3 Operational Use

The most practical finding is that uncertainty can be used as a triage mechanism. If the most uncertain cases are routed to manual review, retained predictions become more reliable. This does not eliminate model risk, but it provides a realistic path for selective automation: high-confidence cases can be handled efficiently, while borderline or unstable cases can be deferred for additional assessment.

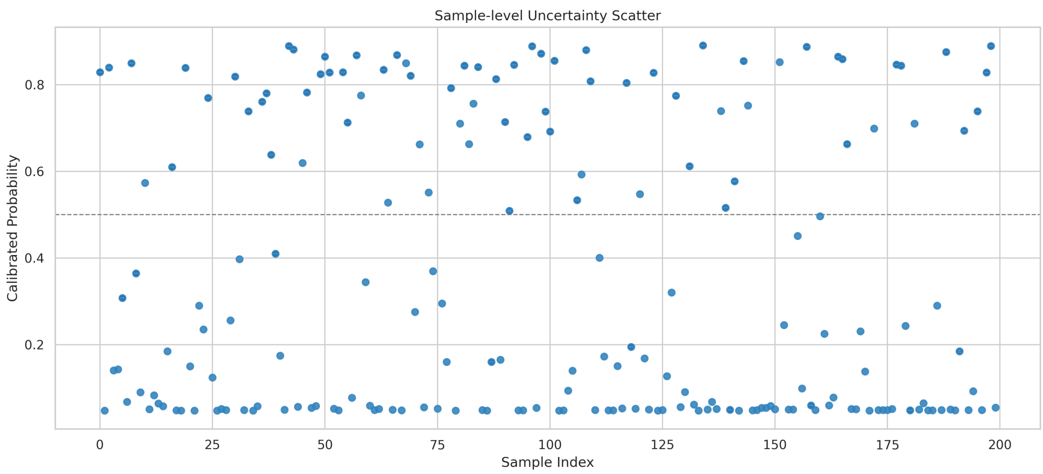

Figure 3: Sample-level calibrated probabilities with one-standard-deviation uncertainty bars. The spread highlights which predictions are locally stable and which ones remain borderline even after calibration.

5 Conclusion

This experiment produced a credible and reasonably well-calibrated smoker-status classifier from structured health-check data. The final model shows strong discrimination, balanced classification quality,

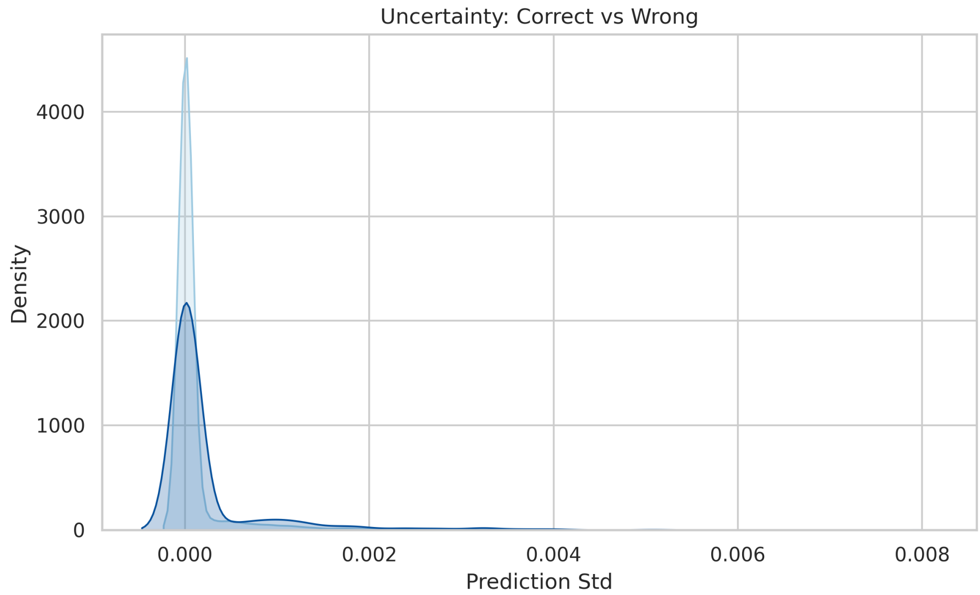

Figure 4: Uncertainty density for correct versus incorrect predictions. The separation between the two distributions indicates that predictive dispersion is informative about error risk.

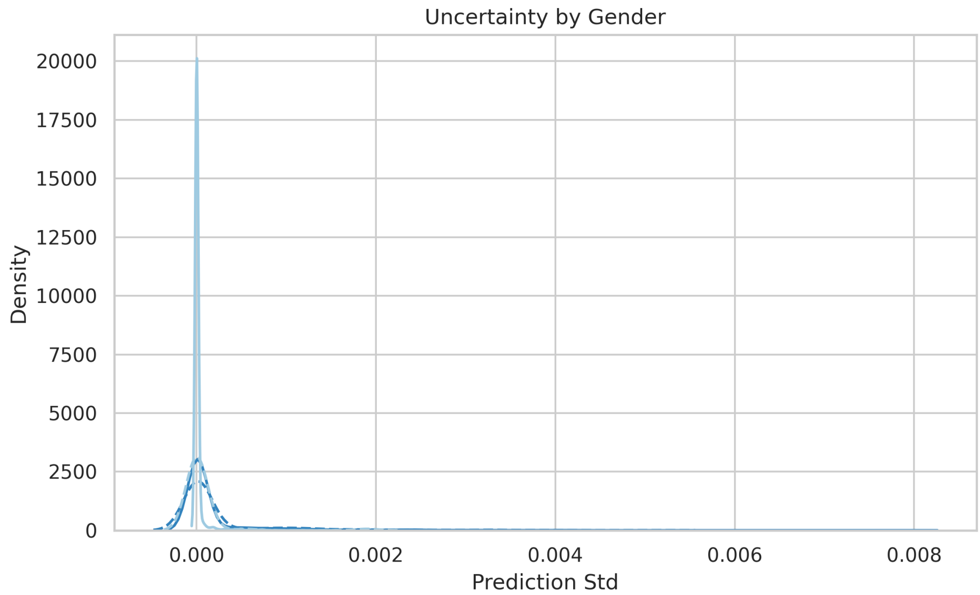

Figure 5: Uncertainty distributions broken down by gender and correctness. This view is useful for checking whether the uncertainty mechanism behaves consistently across key subgroups.

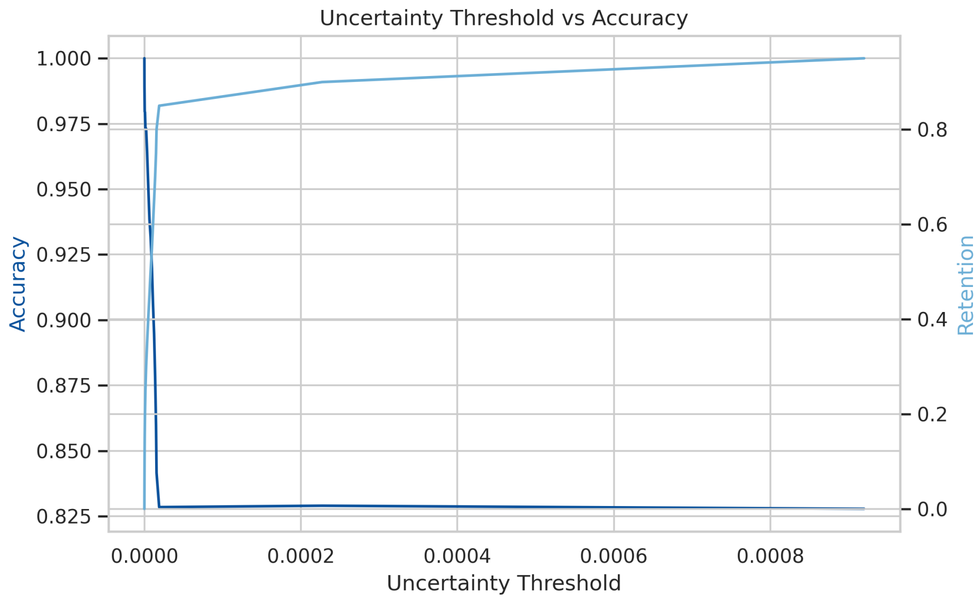

Figure 6: Accuracy-retention tradeoff as high-uncertainty samples are filtered out. The figure illustrates the practical value of selective prediction: reliability improves as the system abstains on more ambiguous cases.

and a useful uncertainty signal. These are meaningful strengths for a screening-style workflow in which the model supports, rather than replaces, expert judgment.

From a deployment perspective, the main takeaway is that the pipeline is most suitable as a calibrated risk stratification tool or a first-pass screening model. It is less appropriate as a fully autonomous decision maker, because the model still depends heavily on strong subgroup signals and because uncertainty, while informative, is not equivalent to a formal abstention guarantee.

Reasonable next improvements would include threshold optimization for specific operating goals, more explicit subgroup calibration analysis, and external validation on a dataset with different smoking prevalence and demographic composition. Those steps would turn this from a strong internal benchmark into a more defensible real-world decision-support system.